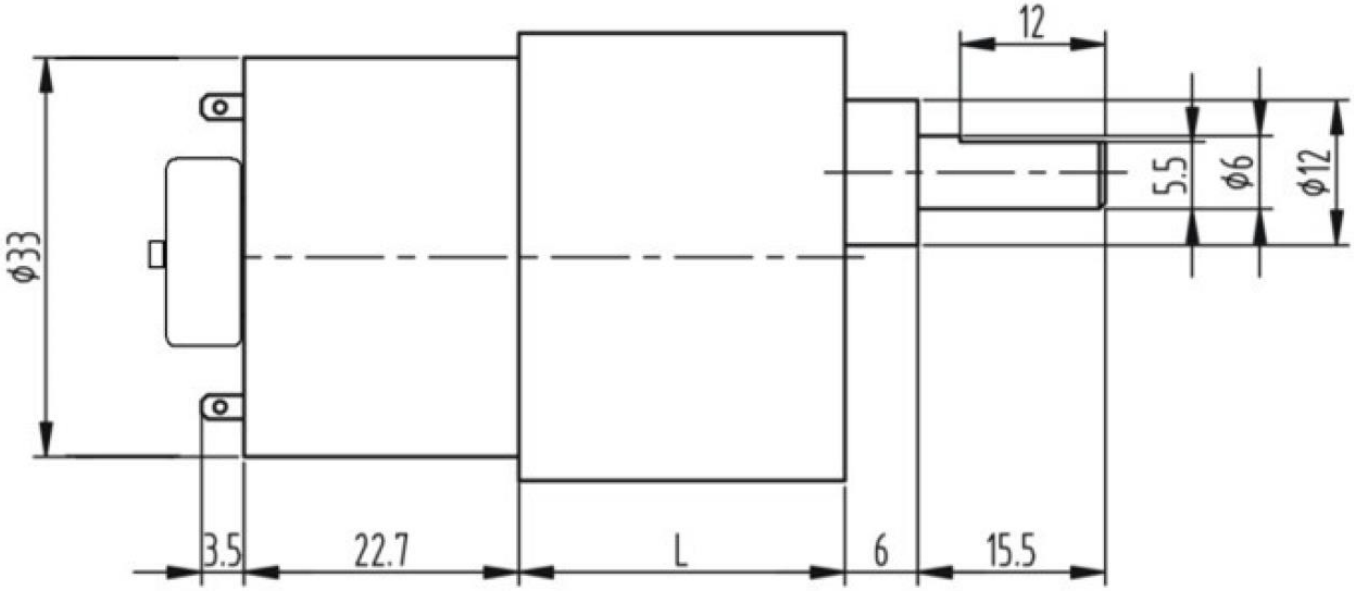
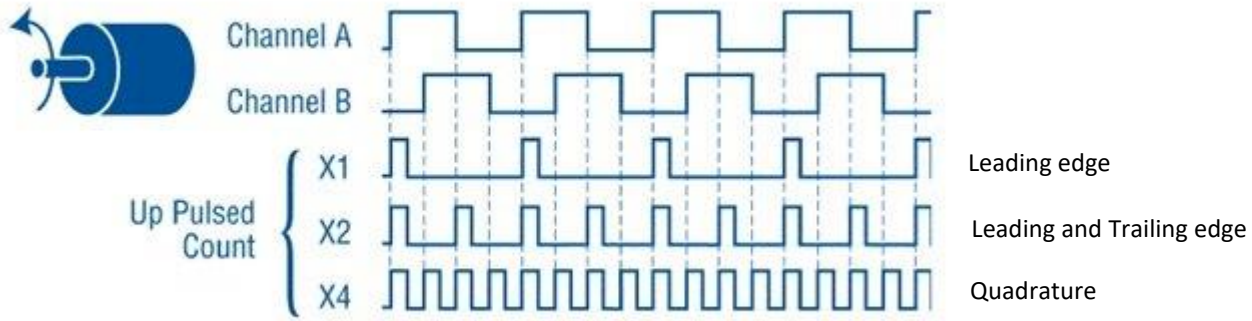
DC Motor: JGB37-520

Voltage		No Load		Load Torque				Stall		Reducer (L)		Weight
Workable	Rated	Speed	Current	Speed	Current	Torque	Output	Torque	Current	Ratio	Size	
Range	Voltage	RPM	mA	RPM	mA	kg·cm	W	kg·cm	A	1:x	mm	g
6-15V	12V	333	120	266	350	1.14	3	4.5	1	30	22	180



Hall Effect Encoder:

Resolution: 11 pulse per revolution (PPR)



Q1 and Q2 in the figure are the hall sensors. Each sensor generates a PWM signal. Every revolution is 11 pulses. Since the motor is attached with a 30x reducer, the encoder generates 30*11=330 pulse per revolution (in Channel A and Channel B).

Servo Motor: TD-8120MG

Specifications:

- Voltage range: 4.8V ~ 6.6V DC
- Dimensions: 40x20x40.5mm (see images for more dimensions)
- No-load speed: 0.18 seconds/60 degrees (4.8V); 0.14 sec/60 degrees (6.6V)
- Torque: 18.5kg · cm (4.8V); 21.8kg · cm (6.6V)
- Rotation range: 180 degrees
- Waterproof (splash water)
- Material gears: Metal

Note: This servo can draw relatively high currents, especially with a load (up to approximately 2500mA at 6V at standstill). So choose a suitable power supply for the servo. It is not recommended to use the Arduino as power supply.

Note: If the servo has less than 180 degrees range, the minimum (500us = 0.5ms) and maximum pulse width (2500us=2.5ms) must be adjusted in the controller.

In the given example, a 50Hz PWM signal is generated by a 8Mhz internal clock, Prescaler is 160, Auto-reload is 1000. Thus, every tick takes 20us.

It is noted that the **HSI clock of STM32F407VET is 16Mhz** but it is 8Mhz in STM32F103VET.

To control the angular position of the servo motor depends on the pulse width.

The minimum pulse width is 25 ticks = 25*20u = 500u and the maximum is 125ticks=2500us

Due to the steering, the working range is roughly between 65 and 85.